

| DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING     |                             |                              |                                                     |     |                      |                                         |
|----------------------------------------------------|-----------------------------|------------------------------|-----------------------------------------------------|-----|----------------------|-----------------------------------------|
| Course Code: 24CS111/24CS211                       |                             |                              | Course Title: C Programming - INTEGRATED LABORATORY |     |                      |                                         |
| Total Credits: 04<br>L-45: T-0: P-30: SL and TW-45 |                             |                              | SDGs addressed: 4, 9                                |     | Course Category: ESC | Course Type: Theory With Integrated Lab |
| Teaching Learning Process                          |                             | Total Contact Hours/semester | Assessment Weightage and Marks                      |     |                      |                                         |
|                                                    | Lecture                     | 45                           |                                                     | CIE | SEE                  | Total                                   |
|                                                    | Tutorial                    | -                            | Weightage, %                                        | 40  | 60                   | 100                                     |
|                                                    | Practical                   | 30                           | Maximum Marks                                       | 40  | 60                   | 100                                     |
|                                                    | Self-Learning and Term Work | 45                           | Minimum Marks                                       | 20  | 25                   | 45*                                     |
|                                                    | Total                       | 120                          |                                                     |     |                      |                                         |

### LIST OF PROGRAMS FOR PRACTICAL SESSIONS

Develop an algorithm/draw a flowchart, write a 'C' program and execute the following:

| Weeks | List of Programs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | No. of Hours |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 1.    | (i) To accept two numbers and perform basic arithmetic operations as follow: +, -, *, /, %<br>(ii) To find the volume of a Sphere using the formula,<br>$\text{Volume} = 4 / 3 \Pi r^3$<br>(iii) To find the area of a Triangle with 3 sides given using formula,<br>$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$<br>where a, b and c are lengths of sides of Triangle, and $s = (a+b+c) / 2$                                                                                                                       | 02           |
| 2.    | (i) Given the values of the variables x, y and z, write a program to rotate their values such that x has the value of y, y has the value of z, and z has the value of x.<br>(ii) To read floating-pointing number and then displays the right-most digit of the integral part of the number.<br>(iii) To read floating-pointing number, separate and displays the integral and decimal part of the given.<br>(iv) To print the size of various data types in 'C' programming language using 'sizeof' operator. | 02           |
| 3.    | (i) Using 'ternary' operator, check whether a given number is Positive or Negative.<br>(ii) Using 'ternary' operator, to check whether a given Year is a Leap Year or Not.<br>(iii) Using 'ternary' operator, find the largest of three numbers.<br>(iv) Using Bitwise-AND (&) operator, check whether a given number is Odd or Even.                                                                                                                                                                          | 02           |
| 4.    | (i) To illustrate the use of increment operator (postfix & prefix).<br>(ii) To illustrate the use of decrement operator (postfix & prefix).<br>(iii) To perform the following using bitwise operators: $c = a \& b$ ; $d = a   b$ ;<br>$e = \sim a$<br>$f = a >> n$ ; $g = a << n$ ; $h = a \wedge b$                                                                                                                                                                                                          | 02           |
| 5.    | (i) To determine whether a given number is 'Odd' or 'Even' and print the message NUMBER IS EVEN or NUMBER IS ODD with and without using 'else' statement.<br>(ii) To determine whether a given number is 'Positive', 'Negative' or 'ZERO' and print the message "NUMBER IS POSITIVE", "NUMBER IS NEGATIVE" or "NUMBER IS ZERO" using nested 'if' statement.<br>(iii) To compute all the roots of a quadratic equation by accepting the non-zero coefficients. Print appropriate messages.                      | 02           |
| 6.    | (i) To generate an electricity bill by accepting meter number of the consumer, number of units consumed and print out the detail charges for the below scenario:<br>An electricity board charges the following rates for the use of electricity:<br>➤ for the first 200 units 80 paise per unit<br>➤ for the next 100 units 90 paise per unit<br>➤ beyond 300 units Rs 1 per unit<br>All users are charged a minimum of Rs. 100 as meter charge. If the total amount is more                                   | 02           |

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|     | <p>than Rs. 400, then an additional surcharge of 15% of total amount is charged.</p> <p>(ii) A Telecommunication department charges the following tariffs for the use of Telephone-Line:</p> <ul style="list-style-type: none"> <li>➤ For the first 100 units 95 paise per unit</li> <li>➤ For the next 75 units Rs. 1.50 per unit</li> <li>➤ Beyond 175 units Rs 2.85 per unit.</li> </ul> <p>All users are charged a minimum of Rs. 75 as service charge. If the total amount is more than Rs 500, then an additional surcharge of 12.5% of total amount is charged. Write a program to read the name of the user, number of units consumed and print out the charges.</p> |           |
| 7.  | <p>(i) To input month number and display its respective month in word.</p> <p>(ii) To simulate a simple calculator to perform the basic arithmetic operations (Consider the operators +, -, x, / and % using 'switch' statement)</p> <p>To check whether a given alphabet is vowel or consonant using 'switch' statement.</p>                                                                                                                                                                                                                                                                                                                                                | <b>02</b> |
| 8.  | <p>(i) To sum Natural, Odd and Even numbers up to 'N'.</p> <p>(ii) To generate and print the first 'N' Fibonacci numbers such that <math>F_n = F_{(n-1)} + F_{(n-2)}</math> where <math>n &gt; 2</math>. A Fibonacci sequence is defined as "the first and second terms in the sequence are 0 and 1. Subsequent terms are found by adding the preceding two terms in the sequence".</p> <p>(iii) To find the sum of individual digits of a positive integer number reducing into single digit.</p> <p>(iv) To reverse a given four-digit integer number and check whether it is a palindrome or not. Output the given number with suitable message.</p>                      | <b>02</b> |
| 9.  | <p>(i) To determine whether a given number is Prime or not.</p> <p>(ii) To generate and print all the prime numbers between range N1 and N2, where 'N1' and 'N2' are value supplied by the user.</p> <p>(iii) To find the value of <math>\cos(x)</math> using the series, <math>1 - x^2/2! + x^4/4! - x^6/6! + x^8/8! - \dots</math> up to N terms accuracy (With and without using in-built function)</p> <p>(iv) To find the value of <math>\sin(x)</math> using the series, <math>x - x^3/3! + x^5/5! - x^7/7! + x^9/9! - \dots</math> up to N terms accuracy (With and without using in-built function)</p>                                                              | <b>02</b> |
| 10. | <p>(i) To input N Real Numbers and to find mean, variance and standard deviation using appropriate formula.</p> <p>(ii) To input N integer numbers into a single dimension array, sort them in to ascending order using "BUBBLE SORT" technique, and then to print both the given array and the sorted array with suitable headings.</p> <p>(iii) To input N integer numbers in into a single dimension array, and then to perform "LINEAR SEARCH" for a given Key integer number and report success or failure in the form of a suitable message.</p>                                                                                                                       | <b>02</b> |
| 11. | <p>(i) To input N integer numbers in ascending order into a single dimension array, and then to perform "BINARY SEARCH" for a given Key integer number and report success or failure in the form of a suitable message.</p> <p>(ii) To perform addition and subtraction of two matrices after checking their compatibility and print both input &amp; output matrices with suitable headings. Use user- defined functions to read and print the matrices.</p>                                                                                                                                                                                                                | <b>02</b> |
| 12. | <p>(i) To read a matrix A (M x N), find the TRANSPOSE of the given matrix and output both the input matrix and the transposed matrix. Use user-defined functions to find the TRANSPOSE.</p> <p>(ii) To find the TRACE and NORM of a given matrix A (M x N) by checking the compatibility and print both input &amp; output matrices with suitable headings. Use user-defined functions to find their TRACE and NORM.</p>                                                                                                                                                                                                                                                     | <b>02</b> |
| 13. | Programs revision from Week-1 to Week-6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>02</b> |
| 14. | Programs revision from Week-7 to Week-12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>02</b> |
| 15. | <b>Laboratory Test</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>02</b> |